Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/11/2007
PI Name:	Banerjee, Rupak K Ph.D.		
Project Title:	Bio-Transport and Metabolism Under Microgravity Supporting Human Physiologic Processes		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cincinnati	**State:**	OH
Zip Code:	45221	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/17/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Sankovic, John M	**Contact Phone:**	216-977-7429
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB46G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Under microgravity conditions, the oxygen transport process attains a higher degree of complexity as the arterial wall structure and biochemical compositions are altered. In addition, the change from earth gravity to microgravity results in changes in blood volume and in the blood volume distribution within the components of the cardiovascular systems. Oxygen transport through arterial vessel walls involves complex flow and mass transfer physics as transport of oxygen needs to be coupled with the carrier blood having pulsatile flow characteristics in microgravity environment. The specific aim of this study is to: 1) analyze the genes from specific cells cultured in the Rotating Wall Vessel (RWV) bioreactor and acquire quantitative information on genetic and biochemical markers of oxygen transport process using the cDNA microarray and enzyme-linked immunosorbent assay (ELISA); and 2) to numerically investigate the simulation of oxygen transfer and distribution; i) under gravity in vascular level with stenosis model; and ii) under micro-g in cellular level with RWV modeling in order to figure out the experimental result and its mechanism.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During three months of the 3rd year of funding (until 09/30/2006), we have cultured rat cardiomyctes (H9c2) in RWV and extracted the nuclear proteins and conducted ELISA for investigating the differential regulation of oxidation sensitive transcription factors. We selected an oxygen-dependent transcription factor: nuclear factor kappa B (NF-kB) for quantifying the regulation of the proteins under simulated micro-g conditions. The limited data that we obtained will be presented at the Summer Bioengineering Conference 2008.
Bibliography Type:	Description: (Last Updated: 09/03/2009)
Articles in Peer-reviewed Journals	Kwon O, Sartor M, Tomlinson CR, Millard RW, Olah ME, Sankovic JM, Banerjee RK. "Effect of simulated microgravity on oxidation-sensitive gene expression in PC12 cells." Advances in Space Research, 2006;38(6):1168-76. http://dx.doi.org/10.1016/j.asr.2006.02.059 , May-2006
Articles in Peer-reviewed Journals	Kwon O, Krishnamoorthy M, Cho YI, Sankovic JM, Banerjee RK. "Effect of blood viscosity on oxygen transport in residual stenosed artery following angioplasty." J Biomech Eng. 2008 Feb;130(1):011003. PubMed PMID: 18298179 , Feb-2008
Articles in Peer-reviewed Journals	Kwon O, Devarakonda SB, Sankovic JM, Banerjee RK. "Oxygen transport and consumption by suspended cells in microgravity: A multiphase analysis." Biotechnol Bioeng. 2008 Jan 1;99(1):99-107. Epub 2007 Jul 5. PMID: 17614322 , Jan-2008
Papers from Meeting Proceedings	Kwon O, Cho YI, Banerjee RK. "Oxygen transport to the avascular wall of a coronary artery stenosis for varying blood viscosity." Summer Bioengineering Conference, Amelia Island, FL, June 2006. Proceedings of the Summer Bioengineering Conference, BIO2006-157567, June 2006. , Jun-2006